Hemanth Vallamkonda · follow-up to my AI Solutions Engineer application · contact

MCP SERVER · COPILOT STUDIO · COST PLANS

An MCP server for cost-plan checks, designed for Copilot Studio

Three read-only tools a Copilot Studio agent can call to take a first pass over a cost plan. Every figure comes back with its basis, its status and its history, and nothing the tools do can approve a number.

A REAL TOOL CALL

`MCP 2026-07-28` over Streamable HTTP

A first pass over a £33.8m synthetic plan: 6 items for review, including a £45,000 transposed line total.

REQUEST `tools/call check_cost_plan`

```
{ "plan_id": "PLAN-OFF-01" }
```

STRUCTURED RESULT (EXCERPT)

```
{
  "checked": {
    "plan_id": "PLAN-OFF-01",
    "line_count": 13,
    "stated_total": 33797891,
    "hash": "11222ce25f70"
  },
  "summary": {
    "items_for_review": 6,
    "checks": 3,
    "reviews": 2,
    "notes": 1
  },
  "flags": [
    "F01 check L01: Line total does not match rate x GIFA",
    "F02 check L03: Inflation may be counted twice",
    "F03 check L09: Percentage base to confirm",
    "F04 review L02: Rate is older than the estimate base date",
    "F05 review L05: Basis too vague to trace",
    "F06 note L04: Outside the synthetic range (soft)"
  ],
  "status": "first_pass"
}
```


Synthetic plan (illustrative), checked against real ONS index values (Open Government Licence). The same code gives this result on the server, in the tests and on this page.

Read-only tools, plain schemas

No approve, edit or send tools. No $ref, no exclusive bounds, no type arrays, string enums, checked on what the server emits; plans go in by reference.

Provenance in every result

Basis, status and history in the structured result, and in the words the model reads.

Tested with more than one client

The `SDK v2` client in current and legacy modes, the `v1` SDK client, raw JSON-RPC with curl, and the MCP Inspector.

TRY IT

An agent's first pass over a synthetic cost plan

Four questions a cost manager might ask, each answered by the tool code the server runs. Change a rate and the results follow. Then the review step, which belongs to a person.

SYNTHETIC DATA (ILLUSTRATIVE) Made by a seeded generator; not RLB, BCIS or market data.

Plan `PLAN-OFF-01` : synthetic office, Stage 2

GIFA 8,400 m², estimate base date Jun 2026. Edit any rate in pounds per m².

REF	ELEMENT	RATE £/m²	TOTAL £
L00	Facilitating works	38	319,200
L01	Substructure ■ F01	285	2,349,000
L02	Superstructure ▫ F04	960	8,064,000
L03	Internal finishes ■ F02	235	1,974,000
L04	Fittings, furnishings and equipment ◌ F06	140	1,176,000
L05	Services ▫ F05	905	7,602,000
L08	External works	150	1,260,000
L09	Main contractor's preliminaries ■ F03	**14%** of groups 1 to 8	3,139,500
L10	Main contractor's overheads and profit	**6%** of groups 0 to 9	1,553,022
L11	Project and design team fees	**9%** of groups 0 to 10	2,469,305
L12	Other project costs	lump sum	180,000
L13	Risks	**7.5%** of groups 0 to 12	2,256,452
L14	Inflation	**4.5%** of groups 0 to 13	1,455,412
Plan total			**33,797,891**

The tool checks PLAN-OFF-01: 13 lines, total £33,797,891, hash `11222ce25f70`.

THE COST MANAGER ASKS

> Check PLAN-OFF-01 before it goes to the client.

THE AGENT CALLS

`tools/call` **`check_cost_plan`**

```
{ "plan_id": "PLAN-OFF-01" }
```

THE TOOL RETURNS

CHECKED PLAN-OFF-01: 13 lines, total £33,797,891, hash `11222ce25f70`

6 items for review

3 to confirm, 2 needing a judgement, 1 soft note; checks first.

TO CONFIRM F01 **L01 Substructure**

Line total does not match rate x GIFA

Stated total £2,349,000; £285.00 x 8,400 m² gives £2,394,000, so the stated figure is £45,000 lower.

Line basis: Measured approximate quantities priced at Jun 2026 (synthetic). Line status: measured estimate.

TO CONFIRM F02 **L03 Internal finishes**

Inflation may be counted twice

This rate carries prices to Jun 2027, 12 months beyond the estimate base date (Jun 2026), and line L14 adds a group 14 inflation allowance of 4.50% from Jun 2026. The overlapping months are priced twice.

Line basis: Synthetic benchmark rate including forecast inflation to Jun 2027. Line status: benchmark.

TO CONFIRM F03 **L09 Main contractor's preliminaries**

Percentage base to confirm

14.00% is applied to groups 1 to 8 (£22,425,000), leaving out group 0 (facilitating works). That is often deliberate, for example when facilitating works are let as a separate contract with their own preliminaries. If it is not intended, the Table 2.1 order (groups 0 to 8, £22,744,200) gives £3,184,188, £44,688 more.

Line basis: Preliminaries at 14 per cent of the works (synthetic assumption). Line status: allowance.

NEEDS A JUDGEMENT F04 **L02 Superstructure**

Rate is older than the estimate base date

Rate dated Dec 2021; the estimate base date is Jun 2026, 54 months later. The index alone (All new work, 121.7 to 150.9) takes £960.00 to £1,190.34 per m², before any location adjustment.

Line basis: Synthetic benchmark set, office median at Dec 2021. Line status: benchmark.

NEEDS A JUDGEMENT F05 **L05 Services**

Basis too vague to trace

The basis reads "TBC", which does not say where the figure came from or when.

Line basis: TBC. Line status: measured estimate.

SOFT NOTE F06 **L04 Fittings, furnishings and equipment**

Outside the synthetic range (soft)

£140.00 per m² is above the synthetic office range for fittings, furnishings and equipment at Jun 2026 (£43 to £94, median £72) by £46 (48.6%). Location not adjusted. Synthetic data: context for a reviewer, not evidence of an error.

Line basis: Client brief allowance for loose furniture, Jun 2026 (synthetic). Line status: allowance.

Cost limit including inflation £33,797,891 (£4,024 per m² GIFA), VAT excluded, from the stated lines.

BASIS Plan PLAN-OFF-01 (data/plans/PLAN-OFF-01.json), 13 lines, hash 11222ce25f70; cost-plan-check rules 1.1; benchmark comparison against the synthetic set SYN-BENCH-2026 at Jun 2026; rebasing suggestions use All new work with location factor 1.00 (not adjusted).

STATUS **First pass** Items for the cost manager to review. The tool has not approved, changed or sent anything, and a line it does not flag is not thereby confirmed.

HISTORY

1. **loaded** Plan PLAN-OFF-01 read from the sample store (data/plans/PLAN-OFF-01.json): 13 lines, stated total £33,797,891, hash 11222ce25f70.
2. **checked** cost-plan-check rules 1.1: arithmetic, basis, base dates, percentage additions, inflation, then the soft benchmark range.
3. **result** 6 items for review (3 checks, 2 reviews, 1 note).

THE COST MANAGER ASKS

Bring the superstructure rate (£960 per m² at Dec 2021) to the estimate base date. No location adjustment at either end.

THE AGENT CALLS

`tools/call` `rebase_rate`


```
{
  "rate": 960,
  "from_month": "2021-12",
  "to_month": "2026-06",
  "series": "all_new_work",
  "from_location_factor": 1,
  "to_location_factor": 1,
  "nrm1_group": 2
}
```


THE TOOL RETURNS

£1,190 per m² GIFA at Jun 2026

£1,190.34 to the penny, from £960.00 at Dec 2021: x 150.9 / 121.7 (All new work), location factors 1.00 to 1.00 (not adjusted).

BASIS Office for National Statistics All new work index (2015=100): Dec 2021 121.7 to Jun 2026 150.9; location factors 1.00 to 1.00 (not adjusted); input £960.00 per m² GIFA at Dec 2021.

STATUS **Indicative** A first-pass figure for the cost manager to review. Not approved, and not a tender price.

HISTORY
1. **input** £960.00 per m² GIFA at Dec 2021, as supplied by the caller for Superstructure.
2. **index ratio** x 150.9 / 121.7 = x 1.239934 (All new work, released 2026-08-13).
3. **location factors** x 1.00 / 1.00 (not adjusted).
4. **result** £1,190.34 per m² GIFA at Jun 2026; displayed as £1,190.

THE COST MANAGER ASKS

> Is the furniture allowance (£140 per m²) reasonable for an office? No location adjustment.

THE AGENT CALLS

`tools/call` `get_benchmark`

```
{
  "building_type": "office",
  "nrm1_group": 4,
  "to_month": "2026-06",
  "location_factor": 1
}
```

THE TOOL RETURNS

£43 to £94 per m² GIFA, median £72

Office, fittings, furnishings and equipment (group 4), Jun 2026; 12 synthetic projects.

Compared on this page: the plan's L04 rate, £140 per m², sits above the range. A prompt to look at the line's basis, not a verdict.

BASIS Synthetic elemental benchmark set (SYN-BENCH-2026, seed 20261104): 12 synthetic projects, low and high are the 10th and 90th percentiles, linear interpolation between order statistics, at Jun 2026; location factor 1.00 (not adjusted).

STATUS **Synthetic** Synthetic, soft range for context only; a cost manager decides what it means for this project.

HISTORY
1. **dataset** Synthetic elemental benchmark set (SYN-BENCH-2026), 12 synthetic office projects, each rebased by the generator to Jun 2026 with All new work.
2. **range** Low and high are the 10th and 90th percentiles, linear interpolation between order statistics: low £43.47, median £71.97, high £94.22 at Jun 2026.
3. **location factor** x 1.00 (not adjusted).

THE COST MANAGER ASKS

> Rebase the 14 per cent preliminaries to Jun 2026 as well.

THE AGENT CALLS

`tools/call` `rebase_rate`

```
{
  "rate": 14,
  "unit": "percent",
  "nrm1_group": 9,
  "from_month": "2021-12",
  "to_month": "2026-06",
  "from_location_factor": 1,
  "to_location_factor": 1
}
```


THE TOOL RETURNS

◇ REFUSED `percentage_addition`

A percentage is not rebased. Percentage additions (preliminaries, overheads and profit, fees, risk, inflation) update themselves when the percentage is applied to the rebased rates (NRM 1 section 2.10).

Next step: Rebase the element rates, then apply the same percentage to the new total.

THE COST MANAGER REVIEWS

Each item from the check, with a decision. The agent cannot make these, and nothing here approves the plan.

■ TO CONFIRM F01 L01: Line total does not match rate x GIFA

Not reviewed yet

■ TO CONFIRM F02 L03: Inflation may be counted twice

Not reviewed yet

■ TO CONFIRM F03 L09: Percentage base to confirm

Not reviewed yet

□ NEEDS A JUDGEMENT F04 L02: Rate is older than the estimate base date

Not reviewed yet

□ NEEDS A JUDGEMENT F05 L05: Basis too vague to trace

Not reviewed yet

SOFT NOTE F06 L04: Outside the synthetic range (soft)

Not reviewed yet

0 of 6 items reviewed by the cost manager on this page. The plan is approved in the cost system by a named person, not by the agent: the server has no approve, edit or send tool.

HOW IT PLUGS IN

From a question in Teams to an answer a cost manager can trace

AGENT

Copilot Studio

Generative orchestration on; SharePoint method notes as knowledge, numbers only from tools.

PROTOCOL

MCP over Streamable HTTP

One stateless `POST /mcp`; a token for testing, Entra ID for production.

TOOLS

This server

Check a plan, rebase a rate, a synthetic benchmark. Read-only.

DATA

Indices and plans

ONS output price indices; synthetic benchmarks and plans. A firm's own index plugs in as a file.

ANSWER

With provenance, then review

Basis, status and history in words. A named cost manager approves, outside the agent.

Security boundary: tools read bundled data and make no network calls; Host, Origin, rate limit and token are checked before a request body is read; approval stays with a person.

Open the build guide (https://growthforge.github.io/applications/rlb-cost-plan-agent/build.html)

BACKGROUND THAT FITS

What I have built before

- At MTest Labs I built a multi-agent test-design tool grounded in the firm's own test library and issue logs, with a conversational assistant over the firm's knowledge. Test design went from an 8-hour day, or 2 to 3 days, to 1 to 3 supervised hours; a tester approves every case, and I trained the team.
- At Atelier di Venice I built dozens of automated workflows across operations, marketing and finance through low-code automation and APIs.
- I have used n8n for over a year, and I build in JavaScript and Python, with SharePoint and Microsoft 365 integration.

CONTACT

Happy to walk through the code

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Build guide (https://growthforge.github.io/applications/rlb-cost-plan-agent/build.html) and the pack (https://growthforge.github.io/applications/rlb-cost-plan-agent/downloads/cost-plan-check-pack.zip)

Prepared by Hemanth Vallamkonda as part of an application to Rider Levett Bucknall. Independent and illustrative; uses public information only.

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